

Feature selection: meditation vs normal

The collinearity problem

Many HRV features are, by definition, functions of others, so the method must not treat them as independent evidence. On the healthy reference:

- $SD1 \equiv RMSSD \equiv SDSD$ ($|r| = 1.00$)
- $LF \% \equiv LF\text{-rel}$, $HF \% \equiv HF\text{-rel}$
- $SDNN \approx SD2$ (0.98); $Mean \approx Median$ IBI (1.00)

The model

L1-regularized logistic regression (meditation-state vs healthy), standardized to the healthy frame, solved by proximal-gradient **FISTA**. L1 keeps one representative of each collinear cluster and drops the rest — the model de-duplicates for us.

Instead of one fit, **stability selection**: **200 resamples**, each drawing meditation *records* and healthy *subjects* as whole groups (windows never leak across a split), classes balanced. We report how often each feature is selected.

Most-stably selected

Feature	stability	dir
LF/HF	77 %	↑
HF peak	70 %	↑
LF rel / LF %	67 %	↑
LF peak	64 %	↑
pNN20	61 %	↑
Total power	52 %	↑
LF power	47 %	↑

Subject-grouped 5-fold CV **AUC** = 0.73 ± 0.27 . The model most stably selects the **~0.1 Hz LF-band** features (+pNN20) — coherent with the slow ~6 breaths/min pacing of Chi & Kundalini meditation. **The LF-band story the poster tells, earned by selection rather than asserted** — but modest: no feature above 77 %, only **11 meditators**; exploratory, not a validated classifier.



HeartRateLab

The data behind the poster. Every number, prior and longitudinal plot that would not fit on the Ao sheet — the exact z-score table, the fitted scoring families, the selection model, and P1's own correlation.

Institutional support



Author

Luis Alberto Barradas Chacón
TU Graz · Med Uni Graz

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[abcsds/HeartRateLab.jl](https://github.com/abcsds/HeartRateLab.jl)



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Supplementary data sheet · Julia 1.11



HeartRateLab

HeartRateLab.jl

The data & receipts behind the poster

Is this new experimental data *normal*?



albertobarradas.com

Luis Alberto Barradas Chacón · TU Graz · Med Uni Graz

Scoring families & goodness of fit

The normative prior fitted to each scored feature (MLE over 61,715 healthy windows), with its parameters and KS p .

Feature	Family	Params	KS p
LF/HF	LogN	0.852, 0.869	7e-9
LF power	Gamma	0.868, 705	5e-118
TINN	Gamma	5.32, 30.9	2e-18
Total power	Gamma	0.946, 1650	2e-125
HRV tri-index	Gamma	5.94, 1.74	1e-14
LF rel	Beta	5.13, 8.02	2e-12
LF %	Gamma	8.66, 4.50	3e-4
cCSI	LogN	6.64, 0.882	4e-5
SD2	Gamma	3.73, 18.7	2e-24
SD2/SD1	LogN	1.14, 0.533	1e-7
SDNN	Gamma	3.79, 14.0	4e-29
SD1-SD2 area	LogN	8.21, 1.01	7e-16
CVI	Normal	4.27, 0.439	7e-16
pNN20	Beta	1.81, 3.30	2e-17
pNN50	Beta	0.321, 3.56	2e-98
rRR	Gamma	4.16, 0.881	3e-172
IBI range	Gamma	3.49, 94.1	4e-14
Max IBI	Normal	953, 197	3e-11
Mean IBI	Normal	780, 147	1e-64
Min IBI	Normal	625, 128	5e-27
Median IBI	Normal	780, 151	1e-77
CVSD	Gamma	2.93, 0.0146	6e-44
RMSSD	Gamma	2.40, 14.2	3e-50
SD1	Gamma	2.40, 10.1	3e-50
SDSD	Gamma	2.40, 14.2	3e-50
LF peak	Normal	0.0642, 0.0183	3e-306
sdann	Gamma	0.873, 27.8	6e-28
HF power	Gamma	0.615, 587	7e-208
Min HR	Normal	100, 21.1	1e-40
Mean HR	Normal	79.7, 15.2	2e-95
Max HR	Normal	65.7, 13.8	9e-68
HF peak	Normal	0.213, 0.0633	1e-111
SD HR	Gamma	3.79, 392	1e-9
HF rel	Beta	1.31, 5.44	3e-40
HF %	Gamma	2.48, 7.84	1e-26

KS-rejected at $n \approx 62k$ — these are convenient scoring families, not fitted laws; read z descriptively. Full 53-feature registry at github.com/abcsds/HeartRateLab.jl.

P1: all 35 scored features

P1's window-median value and its z against each reference; z is a per-window descriptive percentile. Sorted by z vs healthy.

Feature	P1 val	z heal.	z med.
LF/HF	22.8	+2.6	+1.7
LF power	3461	+2.6	+0.9
TINN	320	+1.9	+1.2
Total power	5422	+1.8	+0.6
HRV tri-index	17.1	+1.5	+1.0
LF rel	0.583	+1.4	+0.6
LF %	58.3	+1.4	+0.7
cCSI	2453	+1.3	+1.2
SD2	112	+1.2	+0.7
SD2/SD1	5.54	+1.1	+1.2
SDNN	81.5	+1.1	+0.5
SD1-SD2 area	8010	+0.8	+0.1
CVI	4.61	+0.8	+0.1
pNN20	0.492	+0.7	+2.6
pNN50	0.072	+0.4	+0.2
rRR	3.86	+0.3	-0.4
IBI range	340	+0.2	-0.1
Max IBI	980	+0.1	-0.1
Mean IBI	788	+0.1	+0.1
Min IBI	626	0.0	+0.1
Median IBI	782	0.0	+0.1
CVSD	0.038	0.0	-0.5
RMSSD	29.4	0.0	-0.5
SD1	20.8	0.0	-0.5
SDSD	29.4	0.0	-0.5
LF peak	0.064	0.0	-0.2
sdann	15.2	0.0	-0.1
HF power	143	-0.2	-0.7
Min HR	95.8	-0.2	-0.2
Mean HR	76.1	-0.2	-0.3
Max HR	61.2	-0.3	-0.1
HF peak	0.167	-0.7	-0.5
SD HR	736	-1.0	-0.6
HF rel	0.027	-1.5	-1.1
HF %	2.72	-2.1	-1.5

All 35 scored features; z is a per-window descriptive percentile, not an inferential test. Redundant rows ($SD1 \equiv RMSSD \equiv SDDSD$, $LF\% \equiv LF\text{-rel}$) are handled by the selection model — never read them as independent evidence.

Breathing rate = frequency: why P1's LF is high

A breathing rate maps to a spectral frequency ($\text{Hz} = \text{BPM}/60$); every rate below lands inside the **LF band** (0.04–0.15 Hz), and resonant $\sim 6/\text{min}$ sits at its 0.1 Hz centre.

Breaths/min	Cycle (s)	Freq (Hz)
3.5	17.1	0.058
4.0	15.0	0.067
4.5	13.3	0.075
5.0	12.0	0.083
5.5	10.9	0.092
6.0	10.0	0.100
6.5	9.2	0.108
7.0	8.6	0.117
7.5	8.0	0.125
8.0	7.5	0.133
8.5	7.1	0.142
9.0	6.7	0.150

Bold row = resonant $\sim 6/\text{min} = 0.1\text{Hz}$, the LF-band centre.

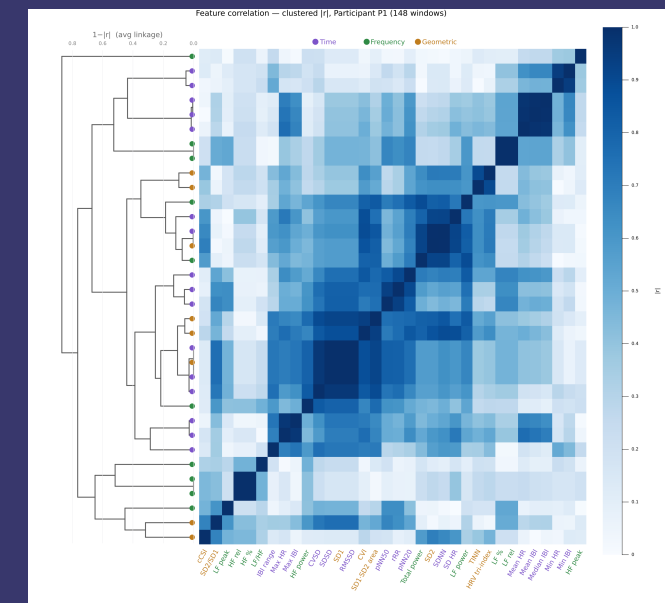
Meditators vs the normal population

The meditation cohort ($n = 11$, during-meditation) scored against the healthy prior — quantile z per feature. **Positive = higher than the general population.**

Feature	z	Feature	z
LF power	+1.2	CVSD	+0.3
LF/HF	+1.0	HF power	+0.3
LF %	+0.9	SD2/SD1	+0.3
LF rel	+0.9	Max IBI	+0.2
Total power	+0.9	Mean IBI	+0.2
CVI	+0.8	Median IBI	+0.2
SD1-SD2 area	+0.8	sdann	+0.2
SDNN	+0.8	Min IBI	+0.1
SD2	+0.7	LF peak	-0.1
rRR	+0.6	Min HR	-0.3
HRV tri-index	+0.6	Mean HR	-0.4
cCSI	+0.4	Max HR	-0.4
TINN	+0.4	pNN20	-0.4
pNN50	+0.4	HF rel	-0.5
SDSD	+0.3	HF peak	-0.7
SD1	+0.3	HF %	-0.7
RMSSD	+0.3	SD HR	-0.8
IBI range	+0.3		

A mild but coherent shift **into the LF band** (LF power +1.2, LF/HF +1.0) with vagal tone barely moved — the same direction as P1, which is why P1 reads **meditator-like**. The meditators are a reference group, not the finding.

P1's own feature correlation



$|r|$ among features across P1's **148 windows**, clustered. The same near-duplicate blocks the population shows (SD1/RMSSD/SDSD, the LF family) — collinearity is **structural**, not a dataset artifact.